Fatemehsadat Mousavinasab, Ph.D.

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U.S. Permanent Resident (no sponsorship required) | Spanish (beginner), English (fluent), Persian (native)

RESEARCH SUMMARY

Immunologist studying how T lymphocytes develop, are tolerized, and go wrong in autoimmunity. My dissertation uses a novel HLA-DQ8 humanized, Foxp3-CNS1 mutant mouse model to define how peripheral regulatory T cells maintain oral tolerance to gluten in celiac disease. I also developed human CD4-expressing murine hybridomas to improve the assessment of peptide-MHC II interactions in humanized mouse models, including gluten peptide presentation by HLA-DQ8. Earlier work examined regulatory T cells and eosinophil biomarkers in eosinophilic GI disease and macrophage polarization in IBD. Eleven publications (6 research articles, 4 review articles, 1 book chapter). Ph.D. expected December 2026 – March 2027.

EDUCATION

Ph.D., Biomedical Sciences, Georgia State University, Atlanta, GA *Expected Dec 2026 – Mar 2027*
GPA 4.19. Focus: T-lymphocyte development and function in celiac disease and systemic autoimmunity. Dissertation: Peripheral regulatory T cells in oral tolerance to gluten in celiac disease. Mentor: Leszek Ignatowicz, Ph.D.

M.Sc., Medical Immunology, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*
GPA 3.86. Thesis: Regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis vs. GERD and controls.

B.Sc., Medical Laboratory Sciences (GPA 3.93), Arak University of Medical Sciences, Iran *2008 – 2012*

RESEARCH AND MENTORSHIP EXPERIENCE

Ph.D. Candidate, Center for Translational Immunology, Georgia State University *2022 – Present*

- Developed a novel HLA-DQ8 humanized, CNS1 mutant (pTreg-deficient) mouse model with a Foxp3-GFP reporter to test the role of peripheral regulatory T cells in oral tolerance to gluten in celiac disease.
- Designed and ran in vivo studies across four groups (CNS wild-type vs. mutant; gluten-free vs. regular diet), plus adoptive-transfer and cholera-toxin induction models.
- Profiled gut T-cell populations (activated $CD4^+$ cells, Tregs, Th17/Treg balance, cytokines) by multicolor flow cytometry (more than 10 colors), histology, and functional assays.
- Established Nur77-GFP $CD4^+$ T-cell hybridomas specific for human class II MHC/peptide complexes (thymoma fusion, HLA-DQ8 dendritic-cell co-culture, peptide testing); identified elevated autoreactive $CD4^+Foxp3^-$ clones in humanized mice.
- Co-authored a published step-by-step protocol (Bio-protocol, 2026) so other laboratories can reproduce the hybridoma method.
- Managed mouse colonies; performed injections, gavage, blood sampling, surgery, and organ isolation.

Head of Immunology & Instructor, GenIran Research Lab, Tehran, Iran *2020 – 2022*

- Led the immunology section and immunoassay-based research projects; quality control manager for ISO 17025.

Researcher, Research Institute for Gastroenterology & Liver Diseases, Tehran, Iran *May – Dec 2019*

- Designed projects on immunologic responses in IBD, including macrophage polarization during acute inflammation and after treatment.

Teaching Assistant, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2017 – 2019*

- Taught practical immunology techniques to master's students, showed them how to operate laboratory instruments, advised their research, and gave exams.

M.Sc. Researcher, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*

- Counted regulatory T cells in esophageal tissue from patients with eosinophilic esophagitis, GERD, and controls; the work was published in Allergologia et Immunopathologia.

PUBLICATIONS

Research articles and methods

- **Mousavinasab F**, Szurek EA, Cebula A, Ignatowicz L, Kuczma MP. Improved protocol for establishing CD4+ hybridomas specific for human class II MHC/peptide complex. *Bio-protocol*. 2026;16(13):e5737. doi:10.21769/BioProtoc.5737.
- Khatami S, Ebrahimi S, Mahjoub FE, Yousefi A, Hosseini AH, Khoshmirsafa M, **Mousavinasab F**, Mansouri M, Shabani M, Mesdaghi M. Detection of eosinophilic cell-free granules based on expression of CCR3 and MBP markers in colonic biopsy of pediatric patients with suspected and confirmed eosinophilic colitis. *Iran J Immunol*. 2026;23(2). doi:10.22034/iji.2026.109789.3152.
- Ebrahimi S, Khatami S, Mahjoub FE, Monajemzadeh M, Yousefi A, Hosseini A, Khoshmirsafa M, **Mousavinasab FS**, Mansouri M, Shabani M, Mesdaghi M. Detection of eosinophilic cell-free granules based on expression of CCR3 and MBP markers in esophageal biopsy specimens of patients with suspected and confirmed eosinophilic esophagitis. *Pediatr Gastroenterol Hepatol Nutr*. 2025;28(1):19-26. PMID: 39839467.

- **Mousavinasab F**, Babaie D, Nilipour Y, Mansouri M, Imanzadeh F, Dara N, Rohani P, Khatami K, Sayyari A, Khoddami M, Kazemiaghdam M, Mesdaghi M. Increased number of regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis in comparison to gastroesophageal reflux disease and control groups. *Allergol Immunopathol*. 47(5):431-436.
- Abdolahy M, Rasouli Sh, Babaie D, Dara N, Imanzadeh F, Sayyari A, Rouhani P, Khatami K, Kazemiaghdam M, Nilipour Y, Khoddami M, Ghadimi F, **Mousavinasab F**, Mesdaghi M. Increased regulatory T cells in peripheral blood of children with eosinophilic esophagitis. *Gastroenterol Hepatol Bed Bench*. 2021 Jan 26:25-30.
- Al-Zaidi HMH, **Mousavinasab F**, Radseresht N, Mirzaei AR, Moradi Y, Mahmoudifar M. Investigation of GJB2 and SLC26A4 genes related to Pendred syndrome genetic deafness patients. *Cell Mol Biomed Rep*. 2023;3(3):163-171.

Review articles and book chapter

- Aghaei-Zarch SM, Nia AHS, Nouri M, **Mousavinasab F**, et al. The impact of particulate matters on apoptosis in various organs: mechanistic and therapeutic perspectives. *Biomed Pharmacother*. 2023;165:115054. PMID: 37379642.
- Hoorzad P, **Mousavinasab F**, Tofigh P, et al. Understanding the lncRNA/miRNA-NFκB regulatory network in diabetes mellitus: from function to clinical translation. *Diabetes Res Clin Pract*. 2023;202:110804. PMID: 37369279.
- **Mousavinasab F**, Karimi R, Taheri S, et al. Microbiome modulation in inflammatory diseases: progress to microbiome genetic engineering. *Cancer Cell Int*. 2023;23:271.
- **Mousavinasab F**, Razani E. Harnessing the potential of vesicular stomatitis virus as a novel therapeutic strategy for cancer treatment. *Iran J Blood Cancer*. 2023;15(2):117-127.
- Salehi-Sardoei A, **Mousavinasab F**, Sayyed RZ, et al. Endophytic fungi: perspectives for microbial engineering (Ch. 11). In: Biostimulants and Protective Biochemical Agents. Elsevier; 2024:167-220. (Book chapter)
- Links to every paper: Google Scholar | Web of Science

PRESENTATIONS

- Digestive Disease Research Day, Emory–Georgia State University (Sep 4, 2026). Investigating the role of peripheral regulatory T cells in oral tolerance to gluten through the development of a novel mouse model of celiac disease. [Dr. Suresh Sitaraman Award, top 3 abstracts]
- AAI Immunology 2026, Boston, MA (Apr 15-19, 2026). Investigating the role of peripheral regulatory T cells in oral tolerance to gluten in celiac disease. [Poster Presentation Award]
- 4th Annual Graduate Conference for Research, Scholarship, and Creative Activity, Atlanta, GA (Feb 11, 2026). Elevated autoreactive $CD4^{+}Foxp3^{-}$ T-cell clones in humanized HLA-DQ8/$Ab^{-/-}$ mice revealed by Nur77-GFP T-cell hybridomas.
- Same mouse model, 2025: Southeast Regional Clinical & Translational Science Conference (Pine Mountain, GA, Mar 5-7); 3rd Annual Graduate Conference for Research, Scholarship, and Creative Activity (Atlanta, Feb 5); 2nd Research Day, Institute for Biomedical Sciences, Georgia State University.

HONORS & AWARDS

- Dr. Suresh Sitaraman Award, top three abstracts, Digestive Disease Research Day, Emory–Georgia State University (Sep 2026); Poster Presentation Award, AAI Immunology 2026, Boston, MA
- Best Oral Presentation, Basic and Advanced Translational Immunology Course, Jaipur, India (Oct 2019); FIMSA travel grant (2019)
- EFIS travel grant, SEEIS 2018, Yerevan, Armenia; EAACI junior member travel grant, Helsinki (2017)

TECHNICAL SKILLS

Immunology & cell biology: Multicolor flow cytometry (FlowJo) and adjusting the compensation matrix, FACS/MACS cell sorting, primary cell culture, dendritic cell generation, T-cell hybridoma generation and peptide testing, Treg suppression assays

Molecular biology & protein: DNA/RNA isolation, PCR, real-time qPCR, ARMS-PCR, RFLP, primer design, cDNA synthesis, gel electrophoresis, ELISA, Western blot, SDS-PAGE, BCA, exosome isolation

Histology & imaging: Tissue processing, H&E staining, immunohistochemistry, immunofluorescence, ImageJ

In vivo models: Mouse colony management, disease induction (celiac, colitis), injections, gavage, surgery, organ isolation

Data & software: GraphPad Prism, SPSS, REST, FlowJo, ImageJ, Adobe Photoshop, EndNote; NCBI/BLAST, genome and protein databases

MEMBERSHIP AND SERVICE

- Vice President, Translational Biomedical Sciences Student Association, Georgia State University.
- Judge, Georgia State Undergraduate Research Conference (April 2024 and April 2025). Peer reviewer, Cellular, Molecular and Biomedical Reports (2023): six manuscripts.
- Member: AAI (American Association of Immunologists), IUIS, EAACI, and FIMSA (Women Speakers).

REFERENCES

Leszek Ignatowicz, Ph.D., Distinguished University Professor, Georgia State University | lignatowicz@gsu.edu | 404-413-6685

Cynthia Nau Cornelissen, Ph.D., Distinguished University Professor, Georgia State University | ccornelissen@gsu.edu | 404-413-6680